`AsmetaComp`: a Tool for Runtime Contract Checking with I/O Abstract State Machines
Soundness Proofs for the Contract Composition Rules of I/O ASMs[*]

1 Setting

We consider the contract-based framework for compositional I/O ASMs introduced in Section 5 of the paper. An I/O ASM contract is a pair (A, G), where A is the assumption and G is the guarantee. Contract satisfaction is expressed in Hoare style as

$$\{A\}\, c\, \{G\},$$

meaning that if component c is executed from a current state in which A holds, and the execution step terminates, then G holds in the resulting next state.

The paper defines the composition formulas recursively, and introduces four binary composition operators together with their contract inference rules:

$$c_1 \mid c_2, \qquad c_1 \parallel c_2, \qquad c_1 <\mid> c_2, \qquad c_1 <\parallel> c_2.$$

The goal of this note is to provide:

1. formal proofs of the soundness of the four contract composition rules;
2. a general soundness theorem for all composition formulas, proved by structural induction.

Convention. We write:

- s for the current state;
- s' for the next state;
- s'' for the next state of s';
- $s \models \varphi$ to mean that formula φ holds in state s.

We also follow the paper in assuming that all the rules hold *modulo invariants*, namely under the assumption that the state invariants of the involved I/O ASMs are preserved.

[*]This artifact refers to the paper "`AsmetaComp`*: a Tool for Runtime Contract Checking with I/O Abstract State Machines*"

2 Auxiliary Lemma

Lemma 1 (Use of component contracts)**.** *Let c be an I/O ASM component satisfying the contract (A, G). Then, for every state s, if*

$$s \models A$$

and the execution step of c from s terminates in state t, then

$$t \models G.$$

Proof. This is exactly the definition of contract satisfaction for an I/O ASM. □

3 Soundness of the Basic Composition Rules

3.1 Sequential composition $c_1 \mid c_2$

Proposition 1 (Soundness of the rule for $c_1 \mid c_2$)**.** *Assume*

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\}, \qquad \{A_{c_2}\}\, c_2\, \{G_{c_2}\}, \qquad G_{c_1} \Rightarrow A_{c_2}.$$

Then

$$\{A_{c_1}\}\, c_1 \mid c_2\, \{G_{c_2}\}.$$

Hence the contract of $c_1 \mid c_2$ is

$$(A_{c_1}, G_{c_2}).$$

Proof. Let s be a state such that

$$s \models A_{c_1}.$$

We must prove that, if the execution of $c_1 \mid c_2$ from s terminates, then the guarantees G_{c_2} are satisfied.

By the operational semantics of the pipe operator, first c_1 is executed and then c_2 is executed using the output of c_1 as input.

Let s' be the state obtained after executing c_1 from s. Since

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\}$$

and $s \models A_{c_1}$, by the previous lemma we obtain

$$s' \models G_{c_1}.$$

By the logical premise

$$G_{c_1} \Rightarrow A_{c_2},$$

it follows that

$$s' \models A_{c_2}.$$

Now let s'' be the state obtained by executing c_2 from s'. Since

$$\{A_{c_2}\}\, c_2\, \{G_{c_2}\}$$

and $s' \models A_{c_2}$, we conclude that

$$s'' \models G_{c_2}.$$

Therefore every terminating execution of $c_1 \mid c_2$ started from a state satisfying A_{c_1} ends in a state satisfying G_{c_2}. Hence

$$\{A_{c_1}\}\, c_1 \mid c_2\, \{G_{c_2}\}.$$

This proves the soundness of the rule. □

3.2 Parallel composition $c_1 \parallel c_2$

Proposition 2 (Soundness of the rule for $c_1 \parallel c_2$)**.** *Assume*

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\}, \qquad \{A_{c_2}\}\, c_2\, \{G_{c_2}\}.$$

Then

$$\{A_{c_1} \wedge A_{c_2}\}\, c_1 \parallel c_2\, \{G_{c_1} \wedge G_{c_2}\}.$$

Hence the contract of $c_1 \parallel c_2$ *is*

$$(A_{c_1} \wedge A_{c_2},\, G_{c_1} \wedge G_{c_2}).$$

Proof. Let s be a state such that

$$s \models A_{c_1} \wedge A_{c_2}.$$

Then

$$s \models A_{c_1} \qquad \text{and} \qquad s \models A_{c_2}.$$

By the semantics of the synchronous parallel composition, c_1 and c_2 are executed on their respective inputs and the resulting outputs are returned together in the next state.

Since

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\}$$

and $s \models A_{c_1}$, the execution of c_1 from s yields the guarantee G_{c_1}.

Similarly, since

$$\{A_{c_2}\}\, c_2\, \{G_{c_2}\}$$

and $s \models A_{c_2}$, the execution of c_2 from s yields the guarantee G_{c_2}.

Therefore, in the next state s' of the parallel composition, both guarantees hold:

$$s' \models G_{c_1} \wedge G_{c_2}.$$

Hence

$$\{A_{c_1} \wedge A_{c_2}\}\, c_1 \parallel c_2\, \{G_{c_1} \wedge G_{c_2}\}.$$

This proves the soundness of the rule. □

3.3 Half-duplex composition $c_1 <|> c_2$

Proposition 3 (Soundness of the rule for $c_1 <|> c_2$)**.** *Assume*

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\}, \qquad \{A_{c_2}\}\, c_2\, \{G_{c_2}\}, \qquad G_{c_1} \Rightarrow A_{c_2}, \qquad G_{c_2} \Rightarrow A'_{c_1},$$

where A'_{c_1} *denotes the assumptions of* c_1 *evaluated in the next state. Then*

$$\{A_{c_1}\}\, c_1 <|> c_2\, \{A'_{c_1}\}.$$

Hence the contract of $c_1 <|> c_2$ *is*

$$(A_{c_1}, A'_{c_1}).$$

Proof. Let s be such that

$$s \models A_{c_1}.$$

We must show that if the execution of $c_1 <|> c_2$ from s terminates, then the next state satisfies A'_{c_1}.

By the semantics of the half-duplex operator, c_1 is executed first, then c_2 is executed using the output of c_1, and finally the output of c_2 is returned as input to c_1 for its *next* future execution.

Let s' be the state obtained after executing c_1 from s. Since

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\}$$

and $s \models A_{c_1}$, we obtain

$$s' \models G_{c_1}.$$

By the premise

$$G_{c_1} \Rightarrow A_{c_2},$$

it follows that

$$s' \models A_{c_2}.$$

Now let s'' be the state obtained after executing c_2 from s'. Since

$$\{A_{c_2}\}\, c_2\, \{G_{c_2}\}$$

and $s' \models A_{c_2}$, we have

$$s'' \models G_{c_2}.$$

Using the last premise,

$$G_{c_2} \Rightarrow A'_{c_1},$$

we derive

$$s'' \models A'_{c_1}.$$

Therefore every terminating execution of $c_1 < | > c_2$ started in a state satisfying A_{c_1} ends in a state satisfying the assumptions required for the next execution of c_1. Hence

$$\{A_{c_1}\}\, c_1 < | > c_2\, \{A'_{c_1}\}.$$

This proves the soundness of the rule. □

3.4 Full-duplex composition $c_1 < || > c_2$

Proposition 4 (Soundness of the rule for $c_1 < || > c_2$)**.** *Assume*

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\}, \qquad \{A_{c_2}\}\, c_2\, \{G_{c_2}\}, \qquad G_{c_1} \Rightarrow A'_{c_2}, \qquad G_{c_2} \Rightarrow A'_{c_1},$$

where A'_{c_1} and A'_{c_2} denote the assumptions of c_1 and c_2 evaluated in the next state. Then

$$\{A_{c_1} \wedge A_{c_2}\}\, c_1 < || > c_2\, \{A'_{c_1} \wedge A'_{c_2}\}.$$

Hence the contract of $c_1 < || > c_2$ is

$$(A_{c_1} \wedge A_{c_2},\, A'_{c_1} \wedge A'_{c_2}).$$

Proof. Let s be a state such that

$$s \models A_{c_1} \wedge A_{c_2}.$$

Then

$$s \models A_{c_1} \qquad \text{and} \qquad s \models A_{c_2}.$$

By the semantics of the full-duplex operator, c_1 and c_2 are executed simultaneously, and their outputs are returned to both components as inputs for their next execution.

Let s' be the resulting next state of the composite execution. Since

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\} \quad \text{and} \quad s \models A_{c_1},$$

we obtain

$$s' \models G_{c_1}.$$

Similarly, since

$$\{A_{c_2}\}\, c_2\, \{G_{c_2}\} \quad \text{and} \quad s \models A_{c_2},$$

we obtain

$$s' \models G_{c_2}.$$

By the compatibility premises,

$$G_{c_1} \Rightarrow A'_{c_2} \qquad \text{and} \qquad G_{c_2} \Rightarrow A'_{c_1},$$

it follows that

$$s' \models A'_{c_2} \qquad \text{and} \qquad s' \models A'_{c_1}.$$

Therefore

$$s' \models A'_{c_1} \wedge A'_{c_2}.$$

Hence

$$\{A_{c_1} \wedge A_{c_2}\}\, c_1 <\,||\,> c_2\, \{A'_{c_1} \wedge A'_{c_2}\}.$$

This proves the soundness of the rule. □

4 General Soundness Theorem

We now extend the previous result from the four primitive binary operators to arbitrary composition formulas.

Definition 1 (Derived contract of a composition formula)**.** Let c be a composition formula generated by the grammar of the paper. The *derived contract* of c, denoted by

$$\mathsf{Ctr}(c) = (A_c, G_c),$$

is defined recursively as follows:

1. if c is an atomic component m, then $\mathsf{Ctr}(m)$ is the given contract of m;
2. if $c = c_1 \mid c_2$, then
$$\mathsf{Ctr}(c) = (A_{c_1}, G_{c_2}),$$
provided the side condition $G_{c_1} \Rightarrow A_{c_2}$ holds;
3. if $c = c_1 \parallel c_2$, then
$$\mathsf{Ctr}(c) = (A_{c_1} \wedge A_{c_2},\, G_{c_1} \wedge G_{c_2});$$
4. if $c = c_1 <\,|\,> c_2$, then
$$\mathsf{Ctr}(c) = (A_{c_1}, A'_{c_1}),$$
provided the side conditions $G_{c_1} \Rightarrow A_{c_2}$ and $G_{c_2} \Rightarrow A'_{c_1}$ hold;
5. if $c = c_1 <\,||\,> c_2$, then
$$\mathsf{Ctr}(c) = (A_{c_1} \wedge A_{c_2},\, A'_{c_1} \wedge A'_{c_2}),$$
provided the side conditions $G_{c_1} \Rightarrow A'_{c_2}$ and $G_{c_2} \Rightarrow A'_{c_1}$ hold.

Theorem 1 (General soundness of contract derivation)**.** *Let c be any well-formed composition formula over I/O ASM components. Assume:*

i. every atomic component occurring in c satisfies its own contract;

ii. for every internal use of a composition operator in c, the corresponding logical side conditions required by the inference rule hold;

iii. the executions considered preserve the invariants of the involved I/O ASMs.

If the derived contract of c is

$$\mathsf{Ctr}(c) = (A_c, G_c),$$

then

$$\{A_c\}\, c\, \{G_c\}.$$

In other words, the contract obtained by recursively applying the composition rules is sound for the whole composition formula.

Proof. We proceed by structural induction on the syntax of the composition formula c.

Base case. Suppose that c is an atomic component m. By assumption, m satisfies its contract. If

$$\mathsf{Ctr}(m) = (A_m, G_m),$$

then by definition of contract satisfaction we immediately obtain

$$\{A_m\}\, m\, \{G_m\}.$$

Thus the statement holds in the base case.

Inductive step. Assume that the statement holds for the immediate subformulas c_1 and c_2, namely:

$$\mathsf{Ctr}(c_1) = (A_{c_1}, G_{c_1}) \implies \{A_{c_1}\}\, c_1\, \{G_{c_1}\},$$
$$\mathsf{Ctr}(c_2) = (A_{c_2}, G_{c_2}) \implies \{A_{c_2}\}\, c_2\, \{G_{c_2}\}.$$

We must prove the statement for a formula c obtained by composing c_1 and c_2 with one of the four operators.

Case 1: $c = c_1 \mid c_2$.
By the recursive definition of derived contract,

$$\mathsf{Ctr}(c) = (A_{c_1}, G_{c_2}),$$

and the side condition

$$G_{c_1} \Rightarrow A_{c_2}$$

holds. By the inductive hypothesis, both

$$\{A_{c_1}\}\, c_1\, \{G_{c_1}\} \quad \text{and} \quad \{A_{c_2}\}\, c_2\, \{G_{c_2}\}$$

hold. Therefore, by Proposition 1, we conclude

$$\{A_{c_1}\}\, c_1 \mid c_2\, \{G_{c_2}\}.$$

Hence

$$\{A_c\}\, c\, \{G_c\}.$$

Case 2: $c = c_1 \parallel c_2$.
By the recursive definition,

$$\mathsf{Ctr}(c) = (A_{c_1} \wedge A_{c_2},\, G_{c_1} \wedge G_{c_2}).$$

By the inductive hypothesis and Proposition 2, we get

$$\{A_{c_1} \wedge A_{c_2}\}\, c_1 \parallel c_2\, \{G_{c_1} \wedge G_{c_2}\}.$$

Thus

$$\{A_c\}\, c\, \{G_c\}.$$

Case 3: $c = c_1 <|> c_2$.
By the recursive definition,

$$\mathsf{Ctr}(c) = (A_{c_1}, A'_{c_1}),$$

and the side conditions

$$G_{c_1} \Rightarrow A_{c_2}, \qquad G_{c_2} \Rightarrow A'_{c_1}$$

hold. By the inductive hypothesis and Proposition 3, we conclude

$$\{A_{c_1}\}\, c_1 <|> c_2\, \{A'_{c_1}\}.$$

Therefore

$$\{A_c\}\, c\, \{G_c\}.$$

Case 4: $c = c_1 <||> c_2$.
By the recursive definition,

$$\mathsf{Ctr}(c) = (A_{c_1} \wedge A_{c_2},\, A'_{c_1} \wedge A'_{c_2}),$$

and the side conditions

$$G_{c_1} \Rightarrow A'_{c_2}, \qquad G_{c_2} \Rightarrow A'_{c_1}$$

hold. By the inductive hypothesis and Proposition 4, we conclude

$$\{A_{c_1} \wedge A_{c_2}\}\, c_1 <||> c_2\, \{A'_{c_1} \wedge A'_{c_2}\}.$$

Hence

$$\{A_c\}\, c\, \{G_c\}.$$

Since all syntactic cases of the grammar of composition formulas are covered, the thesis follows by structural induction. □

5 Concluding Remark

The previous theorem establishes that the contract derivation system is compositional and sound: once contracts are known for atomic I/O ASM components, and the logical compatibility conditions required by the composition rules are satisfied, the contract obtained for any composite formula is correct with respect to the operational semantics of the composition operators.

This is precisely the formal justification behind the claim in the paper that the inference rules for the composition operators are sound.